Week #8: Eigenvalue Problems

Numerical Methods: Mathematical Foundations

June 22, 2025

Overview

Eigenvalue problems represent one of the most important and challenging areas in numerical linear algebra. The computation of eigenvalues and eigenvectors arises in countless applications, from vibration analysis and stability studies to principal component analysis and quantum mechanics. This week we develop the mathematical theory and algorithms for solving these fundamental problems.

Unlike linear systems, eigenvalue problems are inherently nonlinear and typically require iterative methods. We explore both classical methods like the power iteration and modern algorithms like the QR method, analyzing their convergence properties and computational complexity. The mathematical theory reveals deep connections between eigenvalue algorithms and the fundamental properties of matrices.

Our approach emphasizes the mathematical theory underlying eigenvalue computations, eigenvalue sensitivity and perturbation analysis, and the geometric insights that guide algorithm design and implementation.

1 Mathematical Foundation of Eigenvalue Problems

1.1 Basic Definitions and Properties

Definition. Eigenvalue Problem

For a matrix $A \in \mathbb{R}^{n\times n}$, the eigenvalue problem is to find scalars λ (eigenvalues) and nonzero vectors x (eigenvectors) such that:

$$Ax = \lambda x$$

Theorem 1.1 (Characteristic Polynomial)**.** *The eigenvalues of A are the roots of the characteristic polynomial:*

$$p(\lambda) = \det(A - \lambda I)$$

This is a polynomial of degree n, so A has exactly n eigenvalues (counting multiplicities) in $\mathbb{C}$.

Definition. Spectral Properties

For matrix A:

- **Spectrum**: $\sigma(A) = \{\lambda : \det(A - \lambda I) = 0\}$
- **Spectral radius**: $\rho(A) = \max_{\lambda\in\sigma(A)} |\lambda|$
- **Algebraic multiplicity**: Multiplicity of λ as a root of the characteristic polynomial
- **Geometric multiplicity**: $\dim(\ker(A - \lambda I))$

1.2 Special Matrix Classes

Theorem 1.2 (Symmetric Matrices)**.** *For symmetric $A \in \mathbb{R}^{n\times n}$:*

- *All eigenvalues are real*
- *Eigenvectors corresponding to distinct eigenvalues are orthogonal*
- *A is diagonalizable: $A = Q\Lambda Q^T$ where Q is orthogonal*

Theorem 1.3 (Normal Matrices)**.** *A matrix A is normal if $A^TA = AA^T$. Normal matrices satisfy:*

- *A is unitarily diagonalizable: $A = Q\Lambda Q^*$*
- $\|A\|_2 = \rho(A)$
- *The eigenvectors form an orthonormal basis*

2 Power Iteration and Variants

2.1 Basic Power Method

Definition. Power Iteration

Starting with $x^{(0)} \neq 0$, the power iteration generates:

$$x^{(k+1)} = \frac{Ax^{(k)}}{\|Ax^{(k)}\|}$$

Under suitable conditions, this converges to the dominant eigenvector.

Theorem 2.1 (Power Method Convergence)**.** *Assume A has eigenvalues $|\lambda_1| > |\lambda_2| \geq \cdots \geq |\lambda_n|$ and $x^{(0)}$ has a nonzero component in the direction of the dominant eigenvector. Then:*

$$\|x^{(k)} - \pm v_1\| = O\left(\left|\frac{\lambda_2}{\lambda_1}\right|^k\right)$$

where v_1 is the dominant eigenvector.

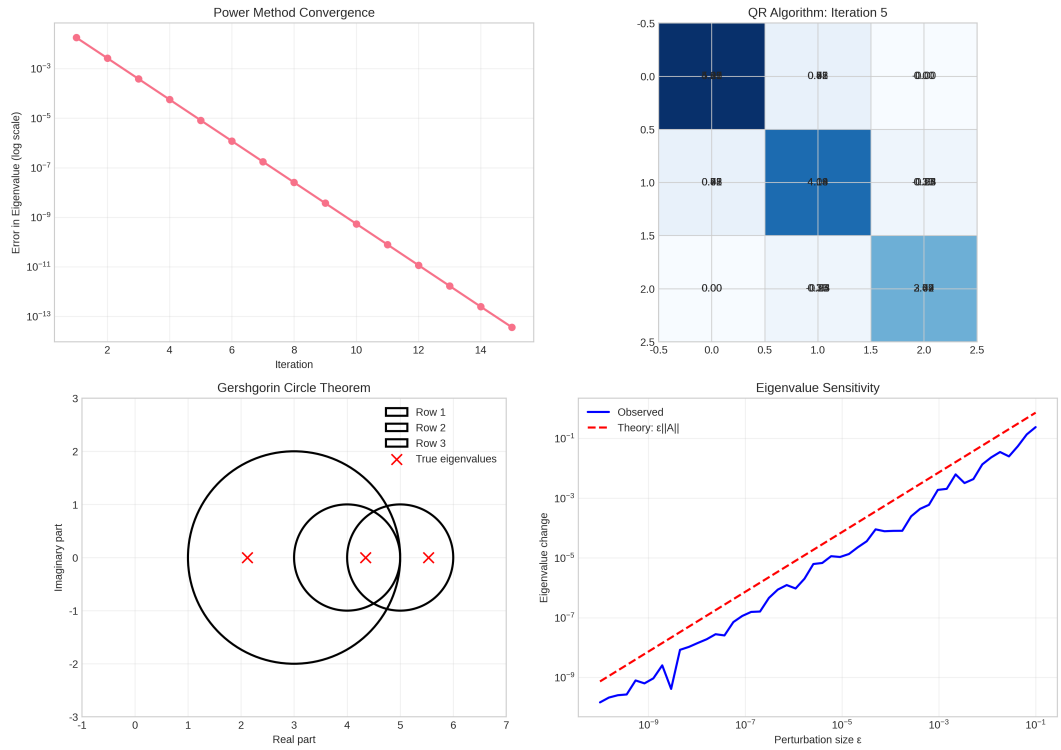

Figure 1: Eigenvalue algorithm comparison. The left panel shows convergence rates of different methods. The right panel illustrates the geometric interpretation of the power method and its convergence to the dominant eigenvector.

2.2 Inverse Iteration

Definition. Inverse Iteration

To find the eigenvalue closest to a shift σ, apply power iteration to $(A - \sigma I)^{-1}$:

$$(A - \sigma I)y^{(k+1)} = x^{(k)}, \quad x^{(k+1)} = \frac{y^{(k+1)}}{\|y^{(k+1)}\|}$$

This converges to the eigenvector corresponding to the eigenvalue closest to σ.

2.3 Rayleigh Quotient Iteration

Definition. Rayleigh Quotient

For nonzero x, the Rayleigh quotient is:

$$\rho(x) = \frac{x^T A x}{x^T x}$$

This provides an approximation to the eigenvalue corresponding to eigenvector x.

Definition. Rayleigh Quotient Iteration

Combine inverse iteration with adaptive shifting:

$$(A - \rho(x^{(k)})I)y^{(k+1)} = x^{(k)}, \quad x^{(k+1)} = \frac{y^{(k+1)}}{\|y^{(k+1)}\|}$$

Theorem 2.2 (Cubic Convergence of Rayleigh Quotient Iteration)**.** *For symmetric matrices, Rayleigh quotient iteration converges cubically to simple eigenvalues.*

3 QR Algorithm

The QR algorithm is the most important method for computing all eigenvalues of a matrix.

3.1 Basic QR Algorithm

Definition. QR Algorithm

Starting with $A^{(0)} = A$, iterate:

$$A^{(k)} = Q^{(k)} R^{(k)} \quad \text{(QR decomposition)} \tag{1}$$

$$A^{(k+1)} = R^{(k)} Q^{(k)} \tag{2}$$

Theorem 3.1 (QR Algorithm Convergence)**.** *Under suitable conditions, $A^{(k)}$ converges to a matrix in Schur form, with eigenvalues appearing on the diagonal.*

Intuition: Why QR Works

The QR algorithm performs simultaneous iteration on all columns of the identity matrix. The similarity transformations preserve eigenvalues while driving the matrix toward triangular form. The convergence rate depends on the separation between eigenvalues.

3.2 Shifted QR Algorithm

Definition. Shifted QR Algorithm

Choose shifts $\sigma^{(k)}$ and iterate:

$$A^{(k)} - \sigma^{(k)} I = Q^{(k)} R^{(k)} \tag{3}$$

$$A^{(k+1)} = R^{(k)} Q^{(k)} + \sigma^{(k)} I \tag{4}$$

Good shift strategies dramatically improve convergence rates.

3.3 Hessenberg Reduction

For efficiency, the QR algorithm is typically applied to Hessenberg form.

Definition. Hessenberg Matrix

A matrix H is upper Hessenberg if $h_{ij} = 0$ for $i > j + 1$.

Theorem 3.2 (Hessenberg Reduction)**.** *Any matrix A can be reduced to Hessenberg form $H = Q^T AQ$ using Householder reflections in $O(n^3)$ operations. The QR algorithm on Hessenberg matrices requires only $O(n^2)$ operations per iteration.*

4 Symmetric Eigenvalue Problems

Symmetric matrices have special properties that enable more efficient algorithms.

4.1 Tridiagonal Reduction

Theorem 4.1 (Tridiagonal Reduction)**.** *Any symmetric matrix can be reduced to tridiagonal form using Householder reflections:*

$$A = QTQ^T$$

where T is symmetric tridiagonal and Q is orthogonal.

4.2 QR Algorithm for Tridiagonal Matrices

The QR algorithm for symmetric tridiagonal matrices has several advantages:

- Preserves tridiagonal structure
- Requires only $O(n)$ operations per iteration
- Guaranteed convergence to diagonal form
- Can be implemented with high accuracy

Theorem 4.2 (Wilkinson Shift)**.** *The Wilkinson shift chooses σ as the eigenvalue of the bottom 2×2 submatrix that is closer to the (n, n) entry. This ensures at least linear convergence and typically gives quadratic convergence.*

5 Perturbation Theory

Understanding eigenvalue sensitivity is crucial for numerical algorithms.

5.1 Eigenvalue Perturbation

Theorem 5.1 (Weyl's Theorem)**.** *For Hermitian matrices A and B with eigenvalues $\alpha_1 \geq \cdots \geq \alpha_n$ and $\beta_1 \geq \cdots \geq \beta_n$ respectively:*

$$|\alpha_i - \beta_i| \leq \|A - B\|_2$$

Theorem 5.2 (Bauer-Fike Theorem)**.** *For diagonalizable $A = X\Lambda X^{-1}$ and any eigenvalue $\tilde{\lambda}$ of $A + E$:*

$$\min_i |\tilde{\lambda} - \lambda_i| \leq \kappa(X)\|E\|$$

where $\kappa(X) = \|X\|\|X^{-1}\|$ is the condition number of the eigenvector matrix.

5.2 Eigenvector Perturbation

Eigenvector sensitivity depends on eigenvalue separation.

Theorem 5.3 (Eigenvector Perturbation Bound)**.** *For simple eigenvalue λ with eigenvector x, the perturbed eigenvector satisfies:*

$$\|\tilde{x} - x\| \leq \frac{\|E\|}{gap(\lambda)} + O(\|E\|^2)$$

where $gap(\lambda) = \min_{i:\lambda_i \neq \lambda} |\lambda - \lambda_i|$.

6 Specialized Methods

6.1 Lanczos Algorithm

For large sparse symmetric matrices, the Lanczos algorithm is highly effective.

Definition. Lanczos Algorithm

Starting with q_1 with $\|q_1\| = 1$, generate orthogonal vectors:

$$\beta_{j+1} q_{j+1} = Aq_j - \alpha_j q_j - \beta_j q_{j-1} \tag{5}$$

$$\alpha_j = q_j^T A q_j \tag{6}$$

$$\beta_{j+1} = \|Aq_j - \alpha_j q_j - \beta_j q_{j-1}\| \tag{7}$$

This generates a tridiagonal matrix T_k whose eigenvalues approximate those of A.

6.2 Arnoldi Algorithm

For nonsymmetric matrices, the Arnoldi algorithm generalizes Lanczos.

Definition. Arnoldi Algorithm

Generate orthonormal basis for Krylov subspace $\mathcal{K}_k(A, q_1)$:

$$AQ_k = Q_{k+1} \bar{H}_k$$

where Q_k has orthonormal columns and $\bar{H}_k$ is upper Hessenberg.

Practice Problems

1. Prove that the power method converges linearly with rate $|\lambda_2/\lambda_1|$ when $|\lambda_1| > |\lambda_2|$.
2. Show that the Rayleigh quotient $\rho(x) = x^T Ax/x^T x$ minimizes $\|Ax - \lambda x\|$ over all λ.
3. Implement the QR algorithm with Wilkinson shifts for symmetric tridiagonal matrices and verify quadratic convergence.
4. Analyze the effect of roundoff error on the Lanczos algorithm and explain the loss of orthogonality phenomenon.
5. Derive the implicit Q theorem and explain its role in the convergence analysis of the QR algorithm.

Solution:

Problem 1: If $x^{(0)} = \sum c_i v_i$ where v_i are eigenvectors, then $A^k x^{(0)} = \sum c_i \lambda_i^k v_i = \lambda_1^k (c_1 v_1 + \sum_{i>1} c_i (\lambda_i/\lambda_1)^k v_i)$. The error is dominated by the $(\lambda_2/\lambda_1)^k$ term.

Problem 2: The Rayleigh quotient is the value of λ that minimizes $\|Ax - \lambda x\|^2$. Taking the derivative with respect to λ and setting to zero gives $\lambda = x^T Ax/x^T x$.

Problem 3: The Wilkinson shift typically gives quadratic convergence because it approximates the eigenvalue very well, making the shifted matrix nearly singular.

Problem 4: Roundoff errors cause loss of orthogonality in the Lanczos vectors. This leads to convergence to the same eigenvalue multiple times. Reorthogonalization or selective reorthogonalization can address this issue.

Problem 5: The implicit Q theorem states that if $Q^T AQ = H$ is unreduced upper Hessenberg and $Qe_1 = \gamma e_1$, then Q and H are uniquely determined up to signs. This explains why different QR

implementations converge to the same result.